

Real-Time Social Media Mining and Trust-Aware Sentiment Analysis
Using Large Language Models for Retail Product Feedback Optimization
(Beyond the Stars)

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1. Broad Area of Work

The broad area of this dissertation is Applied Natural Language Processing using Large Language Models (LLMs) for analysis of public retail customer feedback on social media. The volume of customer voice on public platforms is now beyond manual monitoring. As of January 2024, DataReportal reports 5.04 billion active social media users worldwide [1]; Reddit's S-1 filing (Feb 2024) discloses 73.1 million daily and 267.5 million weekly active uniques [2]; and Sprout Social's 2024 Index reports that 51% of consumers expect a same-day brand response on social [3].

At the same time, fake content is a meaningful contaminant: the World Economic Forum (2024) estimates that fake reviews influence $\approx$ USD 152 billion in online purchases each year, with 4%–15% of reviews flagged as fake across studies [4]. Genuine signal mixed with this noise makes a trust-aware analysis layer necessary, not optional.

The work spans the following sub-areas:

- **Natural Language Processing with LLMs:** sentiment classification and aspect-based opinion mining over short, informal English social-media text using pre-trained LLMs via prompt engineering and few-shot learning [5, 6, 7].
- **Trust / Credibility Filtering:** lightweight post-level credibility scoring using account metadata (age, posting cadence, engagement) combined with an LLM-based "is this a genuine customer post" classifier, drawing on prior work in fake-review and bot detection [8, 9].
- **Retail Operational Feedback:** mapping classified posts to a small, fixed taxonomy of retail-relevant aspects (delivery, product quality, returns, customer support, pricing, app/website experience).
- **Data Engineering:** scheduled ingestion of public posts via official APIs, basic deduplication, and structured storage suitable for aggregation and reporting.

This places the dissertation at the intersection of NLP, social-media analytics, and retail e-commerce, using LLMs as the primary analytical engine.

2. Background

2.1 Why this problem matters

Customers regularly post about their shopping experience on public platforms such as [Reddit](#) and X (Twitter) — covering delivery, returns, product quality, pricing, and customer support. For a large retailer, this public discussion is a useful early signal that often surfaces issues before they appear in internal support tickets.

A few publicly available data points indicate the scale and nature of this signal:

Signal	Public, citable data point	Source
Global social media users	~5.04 billion (Jan 2024)	DataReportal “Digital 2024” [1]
Reddit weekly active uniques	~267.5 million (Q4 2023)	Reddit Inc. S-1 (SEC, Feb 2024) [2]
Same-day response expected on social complaints	51% of consumers	Sprout Social Index 2024 [3]
Estimated impact of fake online reviews	~USD 152 billion in influenced purchases annually	World Economic Forum, 2024 [4]
Share of online reviews flagged as fake (typical range across studies)	~4%–15%	WEF 2024; academic surveys on fake reviews [4, 9]
Walmart store footprint	10,500+ stores in 19 countries	Walmart Corporate Fact Sheet (FY2024) [10]
Walmart workforce	~2.1 million associates worldwide	Walmart Corporate Fact Sheet (FY2024) [10]

Within Walmart, retail-relevant subreddits hold a substantial public discussion footprint. The candidate curated 32 Walmart-relevant communities. A snapshot of public Reddit subscriber counts taken on 05 May 2026 is shown below.

Group	Subreddit	Subscribers (snapshot 05 May 2026)
Walmart core	r/walmart	369,446
Walmart core	r/WalmartEmployees	76,586
Walmart core	r/samsclub	75,200

Walmart core	r/OGPBackroom	33,468
Spark / last-mile	r/Sparkdriver	92,878
Spark / last-mile	r/walmartogp	17,730
Spark / last-mile	r/WalmartSparkDrivers	16,049
Walmart international	r/Flipkart	10,259
Walmart international	r/WalmartCanada	8,618
Walmart pharmacy	r/walmart_RX	9,511

That is roughly 710,000+ subscribers across the Walmart-employee, Sam's Club, Spark-driver, OGP/pickup, pharmacy, and India/Canada subreddits — a meaningful pool of candid retail conversation. Wider retail-context communities sampled for benchmarking (not as Walmart signal) include [r/personalfinance](#) ($\approx 21.7\text{M}$), [r/Frugal](#) ($\approx 6.76\text{M}$), [r/MaliciousCompliance](#) ($\approx 4.35\text{M}$), [r/TalesFromRetail](#) ($\approx 1.80\text{M}$), [r/Costco](#) ($\approx 1.55\text{M}$), [r/ecommerce](#) ($\approx 638\text{k}$), [r/doordash_drivers](#) ($\approx 420\text{k}$), [r/deals](#) ($\approx 258\text{k}$), [r/AmazonPrime](#) ($\approx 230\text{k}$), [r/Target](#) ($\approx 224\text{k}$), [r/AmazonFlexDrivers](#) ($\approx 165\text{k}$), [r/Coupons](#) ($\approx 126\text{k}$), and [r/instacart](#) ($\approx 83\text{k}$). Banned/empty subs ([r/PhonePe](#), [r/Equate](#), [r/walmartspark](#), [r/walmartmexico](#)) are excluded.

2.2 Limitations of the current way of working

In the candidate's current role at Walmart Global Tech (order fulfillment / product platform), public customer pain points are read via ad-hoc browsing of Reddit, app-store pages, and keyword-filter dashboards. Three concrete problems result:

1. **Coverage:** only a small fraction of the public posts is actually read each week.
2. **Noise:** keyword filters surface a lot of irrelevant or low-quality content, including spam and bot posts.
3. **Latency:** by the time a recurring complaint is recognized, it has often already grown.

Rule-based and lexicon-based sentiment systems perform poorly on short, sarcastic, slang-heavy retail posts; surveys report large accuracy gaps versus modern transformer / LLM approaches on noisy social text [5, 7, 11, 12].

2.3 Why LLMs now

Recent LLMs (GPT-4 class, LLaMA-2/3, Mistral, etc.) show strong zero- and few-shot performance on classification, extraction, and summarization over informal text [5, 6, 7, 13]. The practical implication is that a small set of well-designed prompts plus a few few-shot examples can produce a working sentiment + aspect classifier without large-scale labelled training data. Fine-tuning is therefore not required on the core path; it is treated as an optional experiment.

2.4 Conservative comparison: current vs. proposed (target deltas for this dissertation)

The table below states the targets this dissertation aims to validate on a controlled evaluation set — intentionally modest and what one student can rigorously evaluate in a semester.

Dimension	Current ad-hoc approach	Proposed prototype (this work)
Discovery cadence	Manual browsing, weekly	Scheduled ingestion (e.g., hourly) from a single source (Reddit)
Volume processed	~tens of posts/week reviewed	All ingested English posts in scope (target: a few thousand per evaluation window)
Sentiment classification	Subjective, inconsistent	LLM-based, target $\geq 80\%$ accuracy / macro-F1 on a manually labeled held-out sample of ~250–300 posts
Aspect tagging	Not performed	Fixed 6-aspect taxonomy, target ≥ 0.70 macro-F1 on the same held-out sample
Trust filtering	Not performed	Heuristic + LLM credibility score, target ≥ 0.80 ROC-AUC on a small balanced sample of suspected genuine vs. spam/bot posts
Time from post to surfaced summary	Days, ad-hoc	Reproducible reporting at the end of each ingestion cycle

These targets are meant as academic evaluation targets on a curated sample, not as production SLAs.

2.5 Organizational relevance

The dissertation prototypes how LLMs combined with a simple trust filter can turn public retail-related social media into a more reliable, structured view of customer feedback. The prototype uses only public data and is not deployed inside Walmart systems. Value to the employing organisation: (a) a demonstrated working pattern, (b) realistic measurements of how well it works, and (c) honestly documented limitations as a baseline for any future internal effort.

3. Objectives

The objectives of this dissertation are deliberately scoped to what one I can implement and evaluate within the dissertation duration:

1. Survey recent literature on (a) LLM-based sentiment / aspect-based opinion mining over short social text, and (b) fake-review and bot detection / credibility scoring; consolidate into the dissertation.
2. Build a data ingestion pipeline that periodically collects retail-related posts from Reddit via the official Reddit API. Optional extension to X (Twitter) free tier or a public retail-review dataset only if access is reliable.
3. Implement an LLM-based analysis module that, for each ingested post:
 - assigns a sentiment label from {positive, negative, neutral},
 - tags one or more aspects from a fixed retail taxonomy of 6 aspects (delivery, product quality, returns, customer support, pricing, app/website experience).
4. Implement a trust score (0–1) per post combining:
 - simple metadata heuristics (account age, posting frequency, presence of repeated/duplicated text),
 - an LLM-based credibility check (“is this likely a genuine customer post”).
5. Produce structured aggregated outputs that, for a chosen time window, show: top aspects discussed, sentiment distribution per aspect, and representative example posts — with low-trust posts excluded.
6. Evaluate the prototype on a manually labeled sample of approximately 250–300 posts:
 - sentiment classification accuracy / macro-F1,
 - aspect-tagging macro-F1,
 - effect of the trust filter measured as the change in aggregate sentiment distribution before vs. after filtering, and ROC-AUC on a small balanced credibility sample.
7. Document the design, evaluation results, limitations, and recommended next steps in the final dissertation report.

4. Scope of Work

Item	Detail
Primary data source	Public posts from Reddit via the official Reddit API, restricted to a fixed list of retail-relevant subreddits and keywords (see Appendix A).
Optional data extension	X (Twitter) free-tier API or a public retail-review dataset (e.g., Amazon Customer Reviews on AWS Open Data, or

	Yelp Open Dataset) — only if access is reliable.
Pre-processing	Cleaning, English-language filtering, and deduplication.
Sentiment + aspect analysis	6-aspect retail tagging using a pre-trained LLM (API or open-source local) via prompt engineering and few-shot learning.
Optional fine-tuning	Light fine-tuning experiment on an open-source LLM, only if time permits.
Trust score per post	Lightweight 0–1 score combining metadata heuristics and an LLM-based credibility check.
Storage	File-based or lightweight database (e.g., SQLite or local NoSQL) for processed posts and aggregated results.
Reporting	Jupyter notebook by default; optional simple Streamlit dashboard if time permits — showing aggregated sentiment per aspect with example posts and trust-score distribution.
Evaluation	Manually labelled held-out sample of ~250–300 posts: accuracy, macro-F1, and the effect of trust filtering.

5. Plan of Work

Project start date: **26 April 2026**. Dates below are tentative and will be reconfirmed with the Supervisor.

Phases	Start Date – End Date	Work to be done
Outline & Literature Review	25 Apr 2026 – 10 May 2026	Literature review on LLM-based sentiment/aspect analysis and fake-review / credibility detection. Submit dissertation outline.
Design	11 May 2026 – 30 May 2026	Finalize data sources, LLM choice, aspect taxonomy, trust-score design, evaluation plan, and overall system design.
Data Ingestion & Pre-processing	11 May 2026 – 30 May 2026	Implement Reddit ingestion, cleaning, English filtering, deduplication, and storage. Collect a baseline working corpus.
Model Development/ LLM Analysis & Trust Scoring	01 June 2026 – 21 June 2026	Implement sentiment classification, 6-aspect tagging, and the trust-score

		module using prompts plus metadata heuristics.
Aggregation & Reporting	22 June 2026 – 30 June 2026	Aggregated summaries per aspect and time window. Notebook-based reporting and optional simple Streamlit view.
Evaluation	01 July 2026 – 10 July 2026	Manual labeling of ~250–300 posts. Measure accuracy / macro-F1 for sentiment and aspects, and ROC-AUC for the trust filter on a small balanced credibility sample.
Dissertation Review	11 July 2026 – 20 July 2026	Submit draft to Supervisor and Additional Examiner; incorporate feedback.
Final Submission	21 July 2026 – 02 Aug 2026	Final review, formatting per WILP guidelines, and submission.

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Appendix A. Reddit community snapshot (data backing 2.1)

Snapshot taken **05 May 2026** from public Reddit subscriber counts.

Group	Subreddit	Subscribers	Created	Status
Walmart core	r/walmart	369,446	2011-03-31	public
Walmart core	r/WalmartEmployees	76,586	2014-08-07	public
Walmart core	r/samsclub	75,200	2011-07-16	public
Walmart core	r/OGPBackroom	33,468	2019-12-09	public
Walmart core	r/Sams_Club	314	—	dead variant

Walmart core	r/WalmartPlus	298	—	low volume
Spark / last-mile	r/Sparkdriver	92,878	2019-11-22	public
Spark / last-mile	r/walmartogp	17,730	—	public
Spark / last-mile	r/WalmartSparkDrivers	16,049	—	public
Spark / last-mile	r/walmartspark	—	—	banned (404)
Walmart pharmacy	r/walmart_RX	9,511	2019-05-13	public
Walmart international	r/Flipkart	10,259	2021-07-20	public
Walmart international	r/WalmartCanada	8,618	2018-02-04	public
Walmart international	r/walmartmexico	—	—	empty
Walmart international	r/PhonePe	—	—	banned (404)
Walmart private brands	r/GreatValue	84	2019-01-08	restricted
Walmart private brands	r/Equate	—	—	banned (404)
Walmart private brands	r/Onn_ , r/MainstaysHome , r/WalmartCelebrations	—	—	not found / negligible
Retail / customer voice	r/MaliciousCompliance	4,349,691	2016-01-04	public
Retail / customer voice	r/TalesFromRetail	1,796,611	2011-11-09	public
Retail / customer voice	r/CustomerService	51,818	2010-11-19	public
Personal finance /	r/personalfinance	21,674,613	2009-02-08	public

shopping				
Personal finance / shopping	r/Frugal	6,764,364	2008-03-03	public
Deals / shopping	r/ecommerce	637,795	2008-03-16	public
Deals / shopping	r/deals	258,497	2008-06-12	public
Deals / shopping	r/Coupons	126,189	2009-02-14	public
Deals / shopping	r/OnlineShopping	8,989	2008-10-09	restricted
Last-mile competitors	r/doordash_drivers	419,892	2016-10-28	public
Last-mile competitors	r/AmazonFlex Drivers	164,908	2016-07-04	public
Last-mile competitors	r/instacart	83,072	2014-04-03	public
Retail competitors	r/Costco	1,548,744	2010-05-25	public
Retail competitors	r/AmazonPrime	230,002	2012-06-01	public
Retail competitors	r/Target	224,006	2010-12-08	public

7. Particulars of the Supervisor and Additional Examiner

	Supervisor	Additional Examiner
Name	Varunendra Pratap Singh	Suhas Joshi
Qualification	B.Tech – Computer Science & Engineering	Master of Computer Applications (MCA)
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8. Remarks of the Supervisor

The proposed dissertation addresses a relevant and timely problem in large-scale retail e-commerce, where customer voice on public social media has become an important leading signal for product and operational decisions. The use of Large Language Models combined with an explicit trust-aware filter is well motivated, because uncritical consumption of social media content — a meaningful share of which is known to be fake or low-credibility — would otherwise mislead any downstream analytics.

The student has scoped the work tightly: ingestion is restricted to public Reddit data, the analysis taxonomy is fixed to six retail-relevant aspects, evaluation is performed on a manually labelled sample of approximately 250–300 posts, and explicit accuracy / macro-F1 / ROC-AUC targets are stated up front. Items that would expand the work beyond what one student can rigorously evaluate in a semester (multimodal, multi-language, production deployment, full competitor benchmarking, business-KPI claims) are clearly listed as out of scope. The plan of work is realistic and includes literature review, design, prototype implementation, evaluation, and review cycles.

The student demonstrates a good understanding of the problem domain, strong implementation skills, and direct exposure to Walmart's e-commerce which is expected to add practical depth to the work without crossing into use of any internal customer data.

I approve the project as Supervisor.

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SECOND SEMESTER OF ACADEMIC YEAR 2025-2026
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DISSERTATION / PROJECT / PROJECT WORK TITLE	Real-Time Social Media Mining and Trust-Aware Sentiment Analysis Using Large Language Models for Retail Product Feedback Optimization (Beyond the Stars)

Signature of Student	Signature of Supervisor	Signature of Additional Examiner
Name: Vishal Singh	Name: Varunendra Pratap Singh	Name: Suhas Joshi